

R Code and Output for Analyses_for_Table S1

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Table S1

Using the Macrogastrinae benchmark to estimate ACG and global Braconidae, Ichneumonoidea, and Hymenoptera (following the same approach we have used for Insecta). We have added Table S1 to present these estimates, which we discuss briefly in the Main Text.

```
library(SpadeR)
library(dplyr)
library(stringr)
library(pairwiseCI)
#library(remotes)
#install_version("CARE1", "1.1.0")
library(CARE1)

## Load Data

#Macrogastrinae

reared <- read.csv("../Data/Microgastrinae_ACG_reared.csv") %>%
  select(BIN, Frequency)

core <- read.csv("../Data/Microgastrinae_ACG_Core_Malaise.csv") %>%
  select(BIN, Frequency)

peri <- read.csv("../Data/Microgastrinae_ACG_Peripheral_Malaise.csv") %>%
  select(BIN, Frequency)

# All insecta core

insec <- read.csv("../Data/Insecta_ACG_Core_Malaise.csv") %>%
```

```

    select(BIN, Frequency)

#Braconidae

Brac <- read.csv("../Data/Braconidae_ACG_Core_Malaise.csv") %>%
  select(BIN, Frequency)

#Ichneumonoidea

Ichn <- read.csv("../Data/Ichneumonoidea_ACG_Core_Malaise.csv") %>%
  select(BIN, Frequency)

#Hymenoptera

hym <- read.csv("../Data/Hymenoptera_ACG_Core_Malaise.csv") %>%
  select(BIN= BIN, Frequency)

## highlight estimator

estimator <- "    iChao1 (Chiu et al. 2014)"

## adding in the trees

#TA = number of tree species in ACG
TA <- 1200

#Tc = number of tree species in CR

TC <- 2500
#Tw = number of tree species on Earth

## estimated tree species

TW <- 73274
TW_CI <- c(73271, 73276)

```

Approach 1: Calculate insect richness worldwide using the CARE estimator for Microgastrinae

```
### Total number of BINS
```

```
reared %>% nrow()
```

```
[1] 889
```

```
core %>% nrow()
```

```
[1] 388
```

```
peri %>% nrow()
```

```
[1] 576
```

```
###
```

Step 1: Combining Malaise and Reared samples of Microgastrinae and estimating the true richness of ACG Microgastrinae

```
full_micro_table <- reared %>%  
  rename(reared = Frequency) %>%  
  full_join(rename(core, core = Frequency)) %>%  
  full_join(rename(peri, peri = Frequency)) %>%  
  mutate(reared = ifelse(is.na(reared), 0, 1)) %>%  
  mutate(core = ifelse(is.na(core), 0, 1)) %>%  
  mutate(peri = ifelse(is.na(peri), 0, 1))
```

```
Joining with `by = join_by(BIN)`
```

```
Joining with `by = join_by(BIN)`
```

```
head(full_micro_table)
```

	BIN	reared	core	peri
1	BOLD:AAA1266	1	0	0
2	BOLD:AAA1634	1	1	1
3	BOLD:AAA2143	1	0	0
4	BOLD:AAA2511	1	0	0
5	BOLD:AAA2668	1	0	0
6	BOLD:AAA3229	1	0	1

```
nrow(full_micro_table)
```

```
[1] 1414
```

```
# Transform observed ../Data to CARE1 input format
micro_care <- as.record(full_micro_table[, -1])

micro_care
```

```
001 010 011 100 101 110 111
246 132 147 666 114 40 69
```

```
set.seed(100)
# Main step to obtain population size estimates
CARE1.print(micro_care)
```

Warning in ni/nhat: Recycling array of length 1 in vector-array arithmetic is deprecated.
Use c() or as.vector() instead.

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Use c() or as.vector() instead.

(1) NUMBER OF IDENTIFIED CASES IN EACH LIST:

```
  n1  n2  n3
889 388 576
```

(2) ESTIMATES BASED ON ANY PAIR OF SAMPLES:

	Petersen	Chapman	se	cil	ciu
pa12	3165	3146	237	2734	3667
pa13	2798	2790	151	2522	3116
pa23	1035	1033	37	970	1115

Note1: Refer to Seber(1982,pages 59 and 60) for Petersen estimator and Chapman estimators as well as s.e formula.

Note2: A log-transformation is used is used to obtain the confidence interval so that the lower limit is always greater than the number of ascertained. Refer to Chao(1987,Biometrics,43,783-791) for the construction of the confidence interval.

(3) SAMPLE COVERAGE APPROACH:

	M	D	Chat	est	se	cil	ciu
Nhat-0	1414	1066	0.495	2156	66	2038	2296
Nhat	1414	1066	0.495	3405	638	2493	5088
Nhat-1	1414	1066	0.495	2394	97	2221	2604

Warning: The estimated sample coverage(overlapping information)is too low so that Nhat is unstable. Recommend the use of Nhat-0 or Nhat-1.

Parameter estimates:

	u1	u2	u3	r12	r13	r23
Nhat-0	0.41	0.18	0.27	-0.32	-0.23	1.08
Nhat	0.26	0.11	0.17	0.08	0.22	2.29
Nhat-1	0.37	0.16	0.24	-0.24	-0.14	1.31

Definitions for the sample coverage approach:

M: number of individuals ascertained in at least one list.

D: the average of the number of invididuals listed in the combination of any two lists omitting the other one.

$\hat{C}$: sample coverage estimate, see Equation (14) of Chao and Tsay(1998).
est: population size estimate.

se: estimated standard error of the population size estimation based on bootstrap replications.

cil: 95% confidence interval lower limit(using a log-transformation).

ciu: 95% confidence interval upper limit(using a log-transformation).

Nhat: Population size estimate for sufficiently high sample coverage cases, see Equation (20) of Chao and Tsay (1998).

Nhat-1: One-step population size estimate for low sample coverage cases; see Equation (2.21) of Chao et al. (1996). This estimator is suggested for use when the estimated se of Nhat is relatively large.

u1,u2,u3: estimated mean probabilities depending on the estimate of N.

r12,r13,r23 etc.: estimated coefficient of covariation(CCV) depending on the estimate of N.

```
# Print the CARE1 input format

## n1 = reared = 886
## n2 = core = 388
## n3 = peripheral = 576

## Using Nhat-1

Nhat1 <- round(estN(micro_care, method = "LSC", se = TRUE, nboot = 200))

micro_all_estimate <- Nhat1[1]
micro_all_CI <- c(Nhat1[3], Nhat1[4])
micro_all_CI
```

```
[1] 2207 2625
```

Step 2. Estimating the undersampling ratio for Core Microgastrinae.

```
## proportion of core malaise of the ichao total

proportion_core <- nrow(core)/micro_all_estimate

proportion_core
```

```
[1] 0.1620718
```

```
CI_proportion_core <- nrow(core)/micro_all_CI

CI_proportion_core
```

```
[1] 0.1758043 0.1478095
```

Step 3 Calculate total number of insects in ACG

```
## Total number of insecta bins in ACG core malaise
n_insects_core <- nrow(insec)

n_insects_core
```

```
[1] 53945
```

```
## Total number of Braconidae bins in ACG core malaise  
n_brac_core <- nrow(Brac)  
  
n_brac_core
```

```
[1] 2160
```

```
## Total number of Ichneumonoidea bins in ACG core malaise  
n_ichn_core <- nrow(Ichn)  
  
n_ichn_core
```

```
[1] 3402
```

```
## Total number of hymenoptera bins in ACG core malaise  
n_hym_core <- nrow(hym)  
  
n_hym_core
```

```
[1] 13934
```

```
## insecta  
total_insec_acg <- n_insects_core*(1/proportion_core)  
  
total_insec_acg
```

```
[1] 332846.2
```

```
CI_total_insect_acg <- n_insects_core*(1/CI_proportion_core)  
  
n_insects_core*(1/CI_proportion_core)
```

```
[1] 306846.9 364963.0
```

```
n_insects_core*(1/CI_proportion_core)
```

```
[1] 306846.9 364963.0
```

```
CI_total_insect_acg
```

```
[1] 306846.9 364963.0
```

```
## Braconidae  
total_brac_acg <- n_brac_core*(1/proportion_core)  
  
total_brac_acg
```

```
[1] 13327.42
```

```
CI_total_brac_acg <- n_brac_core*(1/CI_proportion_core)  
  
n_brac_core*(1/CI_proportion_core)
```

```
[1] 12286.39 14613.40
```

```
n_brac_core*(1/CI_proportion_core)
```

```
[1] 12286.39 14613.40
```

```
CI_total_brac_acg
```

```
[1] 12286.39 14613.40
```

```
## Ichneumonoidea  
total_ichn_acg <- n_ichn_core*(1/proportion_core)  
  
total_ichn_acg
```

```
[1] 20990.69
```



```
CI_total_ichn_acg <- n_ichn_core*(1/CI_proportion_core)
```

```
n_ichn_core*(1/CI_proportion_core)
```

```
[1] 19351.07 23016.11
```

```
n_ichn_core*(1/CI_proportion_core)
```

```
[1] 19351.07 23016.11
```

```
CI_total_ichn_acg
```

```
[1] 19351.07 23016.11
```

```
## Hymenoptera
```

```
total_hym_acg <- n_hym_core*(1/proportion_core)
```

```
total_hym_acg
```

```
[1] 85974.22
```

```
CI_total_hym_acg <- n_hym_core*(1/CI_proportion_core)
```

```
n_hym_core*(1/CI_proportion_core)
```

```
[1] 79258.60 94269.97
```

```
n_hym_core*(1/CI_proportion_core)
```

```
[1] 79258.60 94269.97
```

```
CI_total_hym_acg
```

```
[1] 79258.60 94269.97
```

```
## compile results
```

```
compiled_results <- data.frame(Taxonomic_group = c('Insecta', "Hymenoptera",  
                                                  "Ichneumonoidea", "Braconidae", "Microgasteridae"),  
                              total_core = c(n_insects_core, n_hym_core,  
                                              n_ichn_core, n_brac_core, nrow(core)),  
                              total_acg = c(total_insec_acg, total_hym_acg,  
                                              total_ichn_acg, total_brac_acg, micro_all_estimated_acg),  
                              CI_low = c(CI_total_insect_acg[1],  
                                           CI_total_hym_acg[1],  
                                           CI_total_ichn_acg[1],  
                                           CI_total_brac_acg[1], micro_all_CI[1]),  
                              CI_high = c(CI_total_insect_acg[2],  
                                             CI_total_hym_acg[2],  
                                             CI_total_ichn_acg[2],  
                                             CI_total_brac_acg[2], micro_all_CI[2]))
```

```
compiled_results
```

	Taxonomic_group	total_core	total_acg	CI_low	CI_high
1	Insecta	53945	332846.21	306846.95	364962.95
2	Hymenoptera	13934	85974.22	79258.60	94269.97
3	Ichneumonoidea	3402	20990.69	19351.07	23016.11
4	Braconidae	2160	13327.42	12286.39	14613.40
5	Microgastrinae	388	2394.00	2207.00	2625.00

Step 4. Estimating global insect richness – based on trees

```
##### Total insects
```

```
print("Total world insects based on estimated tree richness")
```

```
[1] "Total world insects based on estimated tree richness"
```

```
total_insect_world <- (TW*total_insec_acg)/TA  
total_insect_world
```

```
[1] 20324144
```

```
CI_total_insect_world <- (TW*CI_total_insect_acg)/TA
```

```
CI_total_insect_world
```

```
[1] 18736586 22285246
```

```
##### Total Hymenoptera
```

```
print("Total world Hymenoptera based on estimated tree richness")
```

```
[1] "Total world Hymenoptera based on estimated tree richness"
```

```
total_hym_world <- (TW*total_hym_acg)/TA
```

```
total_hym_world
```

```
[1] 5249729
```

```
CI_total_hym_world <- (TW*CI_total_hym_acg)/TA
```

```
CI_total_hym_world
```

```
[1] 4839662 5756282
```

```
##### Total Ichneumonoidea
```

```
print("Total world Ichneumonoidea based on estimated tree richness")
```

```
[1] "Total world Ichneumonoidea based on estimated tree richness"
```

```
total_ichn_world <- (TW*total_ichn_acg)/TA
```

```
total_ichn_world
```

```
[1] 1281727
```

```
CI_total_ichn_world <- (TW*CI_total_ichn_acg)/TA
```

```
CI_total_ichn_world
```

```
[1] 1181608 1405402
```

```
##### Total Braconidae
```

```
print("Total world Braconidae based on estimated tree richness")
```

```
[1] "Total world Braconidae based on estimated tree richness"
```

```
total_brac_world <- (TW*total_brac_acg)/TA  
total_brac_world
```

```
[1] 813794.6
```

```
CI_total_brac_world <- (TW*CI_total_brac_acg)/TA  
CI_total_brac_world
```

```
[1] 750227.6 892318.7
```

```
##### Total Microgastrinae
```

```
print("Total world Microgastrinae based on estimated tree richness")
```

```
[1] "Total world Microgastrinae based on estimated tree richness"
```

```
total_micro_world <- (TW*micro_all_estimate)/TA  
total_micro_world
```

```
[1] 146181.6
```

```
CI_total_micro_world <- (TW*micro_all_CI)/TA  
CI_total_micro_world
```

```
[1] 134763.1 160286.9
```

```
## compile results

compiled_results_all <- bind_cols(compiled_results,
                                   data.frame(total_word =
                                             c(total_insect_world,
                                              total_hym_world,
                                              total_ichn_world,
                                              total_brac_world,
                                              total_micro_world),
                                             CI_w_low = c(CI_total_insect_world[1],
                                                         CI_total_hym_world[1],
                                                         CI_total_ichn_world[1],
                                                         CI_total_brac_world[1],
                                                         CI_total_micro_world[1]),
                                             CI_w_high = c(CI_total_insect_world[2],
                                                           CI_total_hym_world[2],
                                                           CI_total_ichn_world[2],
                                                           CI_total_brac_world[2],
                                                           CI_total_micro_world[2]))))

compiled_results_all %>%
  mutate(across(where(is.numeric), round))
```

	Taxonomic_group	total_core	total_acg	CI_low	CI_high	total_word	CI_w_low
1	Insecta	53945	332846	306847	364963	20324144	18736586
2	Hymenoptera	13934	85974	79259	94270	5249729	4839662
3	Ichneumonoidea	3402	20991	19351	23016	1281727	1181608
4	Braconidae	2160	13327	12286	14613	813795	750228
5	Microgastrinae	388	2394	2207	2625	146182	134763
	CI_w_high						
1		22285246					
2		5756282					
3		1405402					
4		892319					
5		160287					